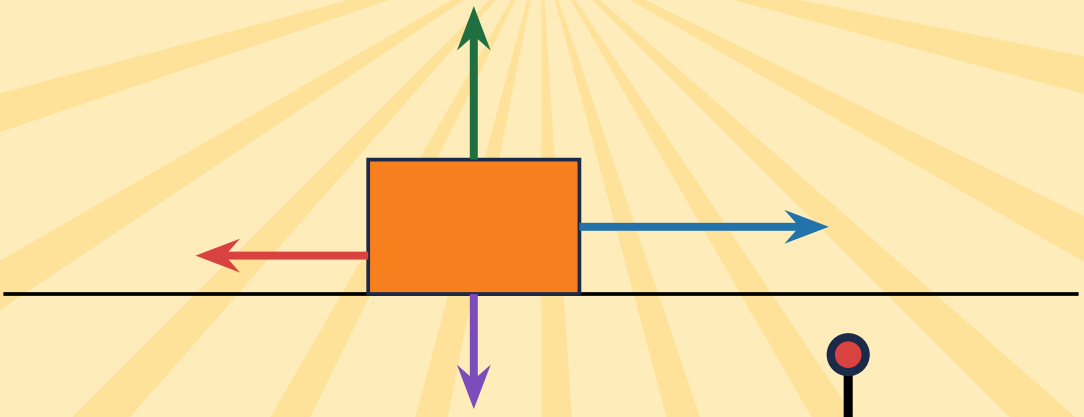
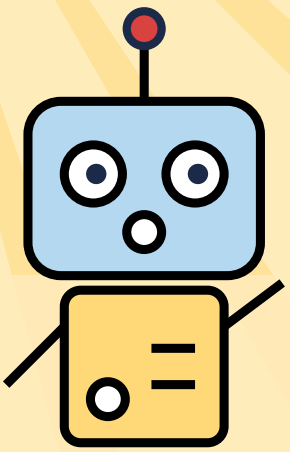
SCIENCE
Book 4

Things That Push Back



Push it. Pull it. Balance it. Draw the arrows.

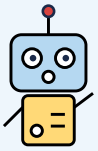
Puzzles, projects and Multix the guide robot



Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

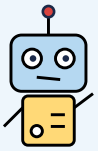


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 45.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I pushed a box across the floor and it stopped. I decided the floor must be broken. Chapter 2 will explain why I should be embarrassed.

In this book

1	Pushes, Pulls and Arrows	6
2	Why Things Stop Sliding	10
3	Falling and Floating	16
4	Balance	21
5	Ramps and Effort	27
6	Finding the Missing Arrow	33
⋆	Slop check	38
⋆	Answers	45
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. Measuring comes from Book 1 and fair testing from Book 2, which you will use again here.

- a small box or block
- rubber bands, all alike
- a ruler marked in mm
- ten identical coins
- a pencil and a notebook
- a board or thick card
- some books
- a toy car
- scrap paper
- a small bag and string

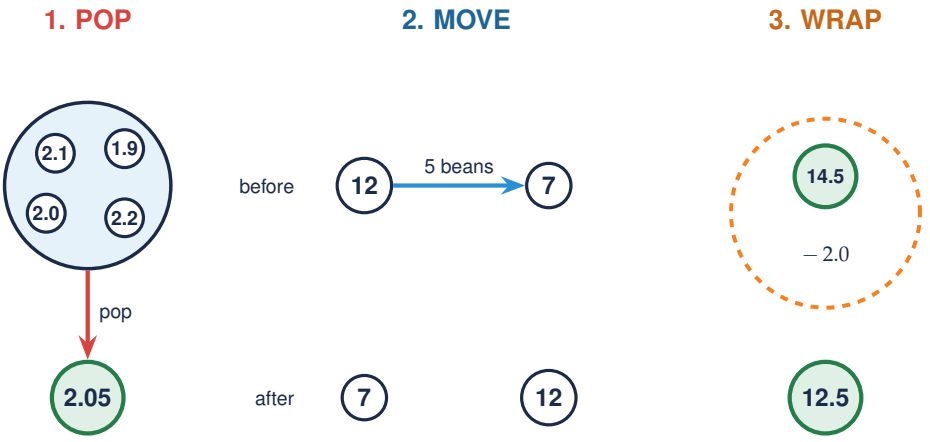
■ **Adult help:** Rubber bands can snap, so keep them pointed away from faces and do not stretch them far. Ramps and boards should rest on something steady. Nothing here needs a flame or mains electricity.

Units in this book

We use the metric system: centimetres (cm), metres (m) and seconds (s). For pushes and pulls we use arrows, and we measure a pull by how far a rubber band stretches. There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

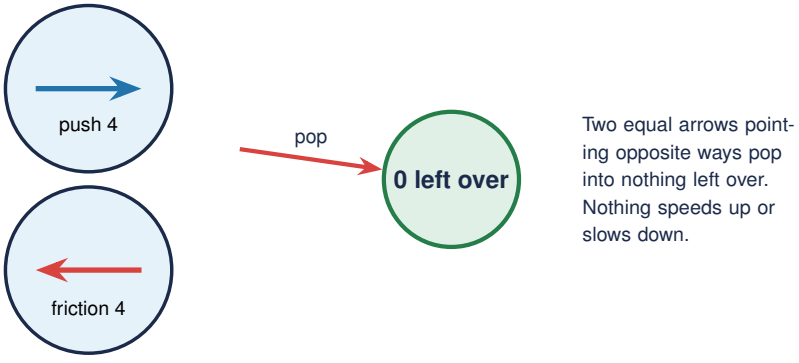


Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

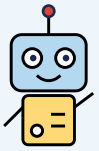
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A force is a circle with an arrow



Two equal arrows pointing opposite ways pop into nothing left over. Nothing speeds up or slows down.



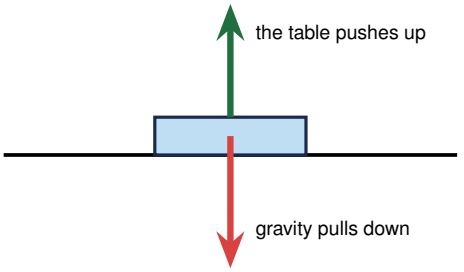
A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 Pushes, Pulls and Arrows

Drawing what acts on a thing

A push or a pull is called a **force**. You use forces all day. You push a door, pull a drawer, kick a ball, lift a bag. A force can make a thing start moving, speed up, slow down, stop, or change direction.

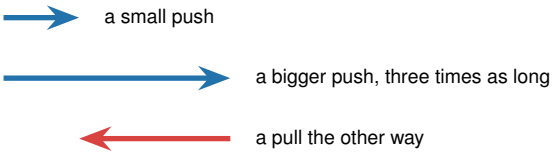
Some forces are easy to see, because a hand does them. Others are harder. A book lying on a desk is not moving. Is nothing acting on it? In fact two forces act on it all the time.



Gravity pulls the book down toward the Earth. The desk pushes the book up. The two arrows are the same length and point opposite ways. They **balance**, so the book stays still.

How to draw a force

A force has two things to tell us: which **direction** it points and how **big** it is. An arrow shows both. The arrow points the way the force acts. A longer arrow means a bigger force.



Try it: Feel the arrows

You need: a book and a table.

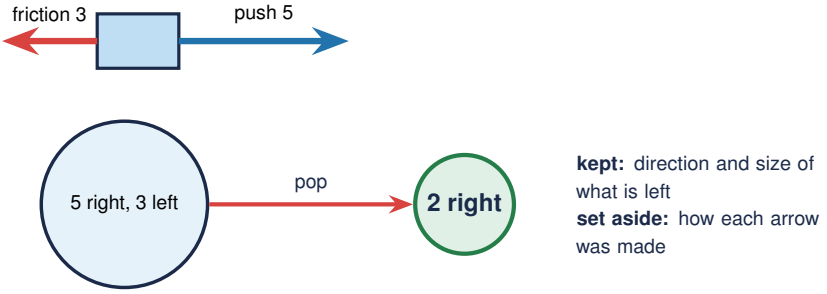
1. Put the book on the table. Press down on it with your hand. Can you feel the table pushing back?

2. Lift the book up a little. Which way is gravity pulling it? Which way is your hand pushing?
3. Push the book along the table. Which way is the push? Is there a second force that pushes back against the slide?

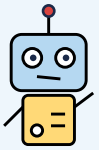
Write what you noticed: _____

Popping the arrows

When more than one force acts on a thing, we want to know what is *left over*. We pop the arrows into one. Arrows on the same line add if they point the same way and subtract if they point opposite ways.



A leftover of 2 to the right means the box will speed up toward the right. A leftover of zero means no change: it stays still, or keeps moving at the same speed.



My box is not moving, so there are no forces acting on it. I will not draw any arrows. That sounds correct to me. Is it?

What did we keep?

A force arrow is a simplification. **We kept** _____ **and we set aside** _____.

Puzzle 1.1: Name the arrows

A bag hangs from a string. Which two forces act on the bag? _____
Are they balanced? _____ How do you know? _____

Puzzle 1.2: Pop them

Find the leftover for each. Use R for right and L for left.

(a) push 6 right, friction 2 left leftover: _____

(b) push 4 right, friction 4 left leftover: _____

(c) push 3 right, wind 5 left leftover: _____

Puzzle 1.3: Which way will it go?

For each, say whether the box speeds up, slows down, or does not change speed, and which way.

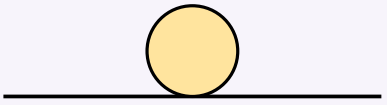
(a) A box sliding right, friction 3 left, no push. _____

(b) A box at rest, push 2 right, friction 2 left. _____

(c) A box at rest, push 5 right, friction 2 left. _____

Puzzle 1.4: Draw the arrows

A ball sits on the floor. Draw the two forces that act on it, with arrows of equal length.



Label each arrow with what is pushing or pulling. _____

Puzzle 1.5: Push or pull?

Write PU (push) or PL (pull).

(a) Opening a drawer by the handle. _____

(b) Closing a door by pressing on it. _____

(c) A magnet drawing a paper clip toward it. _____

(d) Kicking a football. _____

Puzzle 1.6: Make the leftover

A push of size 3 goes right. You want a leftover of 5 to the right. What second arrow do you add, and which way? _____

Puzzle 1.7: One more child

In a tug of war, four children pull left with size 2 each. Four children pull right with size 2 each. Nothing moves. One more child joins the right side with size 1. What is the leftover?

_____

Puzzle 1.8: Sailing

Wind pushes a toy boat right with size 7. The water pulls it left with size 4. What is the leftover, and which way does the boat go? _____

Puzzle 1.9: Your day

Write three pushes and three pulls you did today.

Pushes: _____

Pulls: _____

2 Why Things Stop Sliding

Friction, and how to measure a pull with a rubber band

Push a box across a floor and let go. It slides a little way and stops. Nothing visible stopped it. Something pushed back all along: **friction**.

Friction is a force that acts where two surfaces rub. It always points against the sliding. A box sliding to the right feels friction pointing to the left, so it slows down. Rough surfaces have more friction, and smooth surfaces have less.



A fair test of surfaces

Which surface has the most friction? Use the fair-test habits from Book 2. A good plan changes only the surface.

I change	the surface: a scrap of carpet, smooth card, or a plastic tray
I measure	how far the block slides after it leaves the ramp, nearest cm
I keep the same	the block, the ramp, the point it is let go from, letting go with no push

Try it: Which surface stops the block soonest?

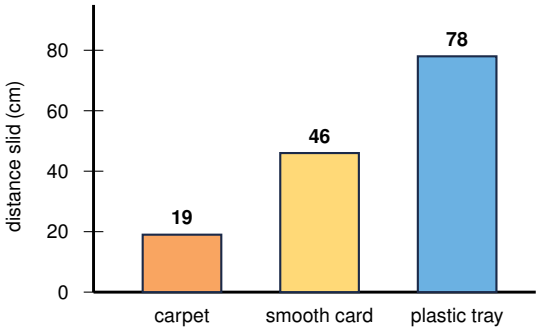
You need: a small block, a smooth board for a ramp, one book to raise it, three surfaces (carpet scrap, smooth card, plastic tray), a ruler or tape measure.

1. Rest the board on one book. Lay a surface flat at the bottom so the block slides onto it.
2. Mark a line on the ramp. Let the block go from that line, with no push.
3. Measure from the bottom of the ramp to the front of the block where it stops.
4. Do three slides for each surface, in a mixed order. Find each average and spread.

	Slide 1 (cm)	Slide 2 (cm)	Slide 3 (cm)	Average (cm)
Carpet				
Smooth card				
Plastic tray				

Surface test (practice numbers, not from a real test)

Carpet: 18, 20, 19 cm. Total = 57. Average $57 \div 3 = 19$ cm. Spread $20 - 18 = 2$ cm.
Smooth card: 46, 44, 48 cm. Total = 138. Average $138 \div 3 = 46$ cm. Spread 4 cm.
Plastic tray: 75, 80, 79 cm. Total = 234. Average $234 \div 3 = 78$ cm. Spread 5 cm.
The gaps between averages, 27 cm and 32 cm, are bigger than any spread.



What makes this correction trustworthy?

What we checked. We changed only the surface, kept the block, ramp and release point the same, and slid the block three times on each surface in a mixed order.

What this check gives us reason to trust. The gaps between surfaces (27 and 32 cm) were far bigger than the wobble within each surface (2 to 5 cm), so the order, carpet then card then tray, is likely real. Changing only the surface means friction from the surface is the likely reason.

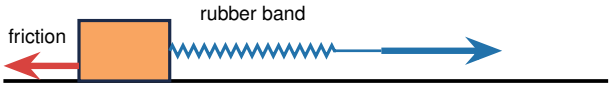
What it does not show. That any carpet, card or plastic behaves like ours. It also does not show how friction works at the tiny scale, and it does not measure the friction itself, only how far the block went.

What did we keep?

We reduced the surface to a distance slid. **We kept** _____ **and we set aside** _____.

Measuring a pull with a rubber band

Distance slid is an indirect measure of friction. To measure friction more directly, pull the block with a rubber band at a steady slow speed. While it moves steadily, your pull and the friction are equal and opposite, so how far the rubber band stretches tells you how big the friction is. A bigger stretch means a bigger pull.



Try it: Measure friction with a rubber band

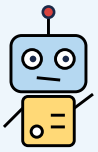
Loop a rubber band over the block. Measure the band's resting length. Pull the block slowly and steadily along each surface. Measure the stretched length while the block is moving. Stretch = stretched length − resting length. Do three pulls on each surface.

	Stretch 1 (cm)	Stretch 2 (cm)	Stretch 3 (cm)	Average (cm)
Carpet				
Smooth card				
Plastic tray				

■ **Adult help:** Rubber bands can snap. Keep the stretch small, point the band away from faces and replace bands that look worn.

Rubber band pulls (practice numbers)

Carpet: 4.6, 5.0, 4.8 cm, average $14.4 \div 3 = 4.8$ cm.
Smooth card: 3.0, 3.3, 3.0 cm, average $9.3 \div 3 = 3.1$ cm.
Plastic tray: 1.8, 2.0, 1.9 cm, average $5.7 \div 3 = 1.9$ cm.
The order matches the slide test: the more it stretches the band, the less far the block slides.



My rubber band stretched 3 cm when I pulled. So the friction is exactly 3 cm. Is 3 cm a force?

More weight, more friction

Stack a second identical block on the first and pull again. Heavier things press harder on the surface, and the friction is bigger.

Stacks of blocks on smooth card (practice numbers)

1 block: 3.1 cm. 2 blocks: 6.0 cm. 3 blocks: 9.2 cm.
Each extra block adds about 3 cm of stretch. Double the blocks gives about double the pull.

Puzzle 2.1: Average the slides

A block slid 22, 25, 23 cm on rug and 62, 58, 60 cm on a wooden floor.
(a) Rug: total _____ average _____ spread _____
(b) Floor: total _____ average _____ spread _____
(c) Which surface has more friction? _____

Puzzle 2.2: Choose the surface

A sledge has to stop soonest at the bottom of a hill. Rank the surfaces from most friction (1) to least: ice _____ grass _____ snow _____ gravel _____

Puzzle 2.3: Spot the unfair test

Lena tests carpet and a tray. She lets the block go from higher up the ramp for the tray. The block slides farther on the tray.
(a) What else changed besides the surface? _____
(b) Can she say the tray has less friction? _____

Puzzle 2.4: Friction helps

Name two things that work only because of friction. Hint: shoes, brakes, matches. _____

Puzzle 2.5: Steady pull

A box is pulled at steady speed with a pull of 4. How big is friction, and which way does it point? _____
If the pull rises to 6, what is the leftover and what happens? _____

Puzzle 2.6: More friction

Pulling the block on card took a 3.1 cm stretch. On the plastic tray it took 1.9 cm. Which surface has more friction? _____
By how much did the stretch differ? _____ cm

Puzzle 2.7: Predict the stack

One block on card needed a 3.1 cm stretch, two blocks 6.0 cm and three blocks 9.2 cm.
Guess the stretch for four blocks. _____ cm
Why is it only a guess? _____

Puzzle 2.8: Which way?

A block slides to the right across a table. Which way does friction point? _____
Now the block slides to the left. Which way does friction point? _____

Puzzle 2.9: Useful or a nuisance

Write two times friction is useful and two times it is a nuisance.
Useful: _____

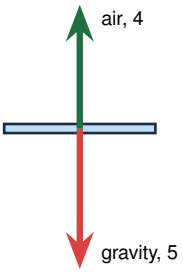
Nuisance: _____

3 Falling and Floating

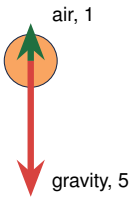
Gravity, air and water push on everything

Let go of a book and it falls. Let go of a sheet of paper and it drifts down slowly, wobbling. Both are pulled down by gravity. So why is the paper so slow?

The answer is that something else acts on the paper. As a thing falls through air, the air pushes up against it. The more surface the thing spreads out, the bigger that upward push. Flat paper spreads a lot of surface, so the air pushes up on it almost as hard as gravity pulls down.



flat paper: leftover 1 down



crumpled paper: leftover 4 down

When the leftover is small, the paper speeds up only a little and falls slowly. When the leftover is big, the thing speeds up quickly. Squash the paper into a ball and the air pushes up much less.

Try it: Flat or crumpled?

You need: two sheets of the same scrap paper, a timer showing tenths of a second, a helper if you can.

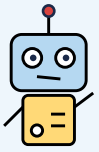
1. Hold one flat sheet as high as you safely reach and let it go. Time the fall to the floor.
2. Crumple the other sheet into a tight ball. Drop it from the same height and time it.
3. Do five drops of each, in a mixed order. Find the average time and spread for each.

	1	2	3	4	5	Average (s)
Flat						
Crumpled						

■ **Adult help:** Do the drops on the floor, not from furniture, with nobody underneath.

Paper drops (practice numbers)

Flat: the ten drops from Book 1 averaged 2.0 s, with a spread of 0.6 s.
Crumpled: 0.7, 0.6, 0.7, 0.7, 0.6 s. Total $= 3.3$. Average $3.3 \div 5 = 0.66$, about 0.7 s. Spread 0.1 s.
The gap between averages is $2.0 - 0.7 = 1.3$ s, much bigger than either spread.



Gravity pulls heavy things harder, so heavy things must always fall faster. I will drop a book and a ball of paper and write that the book wins. Is the book guaranteed to win?

Do heavy things fall faster?

Drop a book and a crumpled ball of paper together from the same height, side by side. Listen. Do you hear one thump or two?

Book and paper ball (practice description)

Dropped together from shoulder height, the book and the ball of paper landed so close together that we heard one thump. With a flat sheet instead of the ball, the book landed long before the paper. Spread-out paper catches the air, and the ball does not.

The upward push of the air is small for the ball and the book, so gravity does almost all the work, and gravity pulls both down with the same speeding-up. The test cannot show gravity is exactly equal, only that any difference is too small to hear.

What makes this correction trustworthy?

What we checked. We dropped flat and crumpled paper from the same height, five times each, and compared the average fall times and the spread.

What this check gives us reason to trust. The gap, 1.3 s, was much bigger than either spread, so the shape of the paper made a real difference to the fall. Changing only the shape (same sheet of paper) points to the air as the cause.

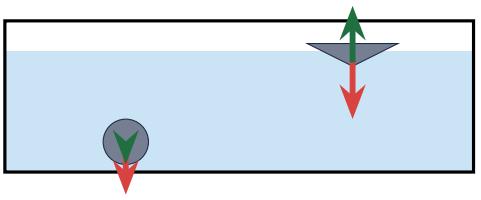
What it does not show. That air is the only reason, or that other shapes follow the same pattern. A stopwatch read to a tenth of a second cannot show tiny differences. The one-thump test is not a measurement of gravity.

What did we keep?

Each arrow keeps the direction and size of a force. **We kept** _____ **and we set aside** _____.

Floating

Water pushes up, too. A piece of clay in a ball sinks. Shape the same clay into a boat and it floats. The clay has not changed. What changed is the shape, which makes the water push up harder.



ball: gravity bigger, sinks boat: water push equals gravity, floats

Try it: Clay boats

Make two clay boats from the same amount of clay: one wide and shallow, one narrow and deep. Float each on water. Add coins one at a time until it sinks. How many coins did each hold?

Wide boat: _____ coins Narrow boat: _____ coins

■ **Adult help:** Clean up spilled water straight away, and keep coins and clay away from small children.

Clay boat results (practice numbers)

Same lump of clay for both. Wide, shallow boat: held 12 coins before sinking. Narrow, deep boat: held 6 coins. The wide boat pushed away more water, so the water pushed up on it harder.

Puzzle 3.1: Time the drops

A crumpled ball took 0.6, 0.7, 0.6, 0.8, 0.6 s to fall. A flat sheet took 1.9, 2.2, 2.0, 2.1, 1.8 s.
(a) Crumpled: total _____ average _____ (b) Flat: total _____ average _____
(c) Gap: _____ s (d) Is the gap bigger than both spreads? _____

Puzzle 3.2: Draw the arrows

A leaf floats down. Draw gravity and the air push, with the air arrow a bit shorter than gravity.



Is the leaf speeding up or slowing down? _____

Puzzle 3.3: Sink or float?

For each, say what the pushes are and whether it sinks or floats: (a) a big wooden plank; (b) a coin; (c) a steel ship. Write a short reason for each.

_____

_____

Puzzle 3.4: A fair test

Marco tests whether paper shape changes fall time. He uses a big flat sheet and a small crumpled sheet from different drops. What else is different? _____
How would you fix it? _____

Puzzle 3.5: How much longer?

Flat paper took about 2.0 s to fall. Crumpled paper took about 0.7 s. How many seconds longer did the flat paper take? _____ s
About how many times as long? _____

Puzzle 3.6: A parachute

A toy parachute has a bigger canopy than a toy without one. The air pushes up on it with a bigger arrow. What will happen to the fall time? _____
Why? _____

Puzzle 3.7: Wide and narrow boats

The wide clay boat held 12 coins. The narrow boat held 6. How many times as many coins did the wide boat hold? _____
Why might a wider boat hold more? _____

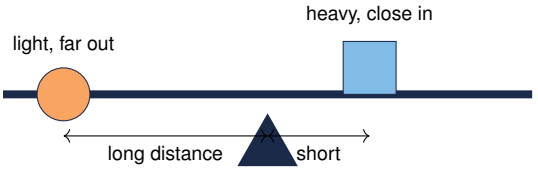
Puzzle 3.8: A speeding ball

A ball is let go. At first gravity pulls down with size 5 and the air pushes up with size 1. What is the leftover? _____
Later, the ball falls at a steady speed. How big is the air's push now? _____

4 Balance

Why a small weight far out can balance a big one close in

Two children on a seesaw can balance even when one is much heavier. The heavier child just sits closer to the middle. A weight does not only matter because of how heavy it is. It also matters where it is.



Try it: A ruler seesaw

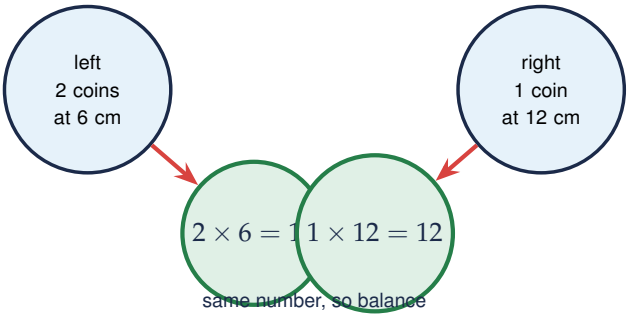
You need: a 30 cm ruler, a pencil, ten identical coins, a bit of tape.

1. Lay the pencil flat on a table. Rest the ruler across it at the 15 cm mark. Adjust until the empty ruler balances.
2. Put two coins on the ruler 6 cm to the left of the middle (the 9 cm mark).
3. Put one coin on the right. Slide it until the ruler balances. Measure its distance from the middle.
4. Try three coins at 4 cm on the left. Where does one coin balance them?

Left: coins	Left: distance (cm)	Right: coins	Right: distance (cm)
2	6	1	
3	4	1	

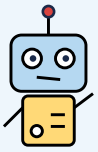
A rule for balancing

Multiply the number of coins by the distance from the middle on each side. If the two numbers match, the ruler balances. If they do not match, the bigger side goes down.



Balancing coins (practice numbers, not from a real test)

Two coins at 6 cm give $2 \times 6 = 12$. One coin balanced them at 12 cm, because $1 \times 12 = 12$.
When the single coin was moved to 10 cm and 14 cm the ruler tipped.
Three coins at 4 cm give $3 \times 4 = 12$. Again one coin balanced them at 12 cm.
Prediction check: the rule said 12 cm both times. The measured balance was within 1 cm of 12.



Three coins on the left at 4 cm, and one coin on the right at 4 cm. Both sides have the same distance, so it must balance. Right?

What makes this correction trustworthy?

What we checked. We predicted the balance point with a rule, then slid the single coin along the ruler and found where the ruler balanced, for two different set-ups.

What this check gives us reason to trust. The measured balance points matched the predictions to within 1 cm both times, so the rule, coins times distance, is working.

What it does not show. That the rule works with big weights, with a bent ruler or with a heavy ruler that is not centred. We used identical coins and one ruler.

What did we keep?

We reduced a seesaw to coins times distance. **We kept** _____ **and we set aside** _____.

Finding the balance point

Every object has a spot where it balances: its **balance point**. A pencil balances in the middle. A spoon balances closer to the bowl, because the bowl is heavier.

Try it: Find the balance point

Rest a pencil, a spoon and a ruler across your finger or the edge of a book. Slide until each balances. Mark the spot with a pencil dot and measure from one end.

Object	Length (cm)	Balance point from one end (cm)
Pencil		
Spoon		
Ruler		

Practice numbers

A 19 cm pencil balanced at 9.5 cm, the middle. A 17 cm spoon balanced at 6 cm from the end of the bowl. A 30 cm ruler balanced at 15 cm.

Puzzle 4.1: Predict the balance

Left: 2 coins at 9 cm. Right: 3 coins. Where must the right coins sit to balance? _____ cm

Puzzle 4.2: Which way?

Left: 3 coins at 5 cm. Right: 2 coins at 8 cm. Which side goes down? _____
Show the numbers: left _____ right _____

Puzzle 4.3: Mystery weight

A mystery bag sits 5 cm to the left of the middle. It is balanced by 2 coins 10 cm to the right. How many coins does the bag weigh? _____

Puzzle 4.4: On the seesaw

A child of weight 30 sits 2 m from the middle. Where must a child of weight 40 sit to balance? _____ m

Puzzle 4.5: Check the rule

A student says the rule is "coins plus distance". Test it: 2 coins at 6 and 1 coin at 12 balance. Does $2 + 6$ equal $1 + 12$? _____ What does that tell you? _____

Predict, then test

Before you test, work out where one coin should sit. Write your prediction, then test it, then write what happened.

Left set-up	Predicted right distance (cm)	Measured (cm)	Off by (cm)
4 coins at 3 cm			
2 coins at 6 cm			
3 coins at 5 cm			

Practice numbers

Left 4 coins at 3 cm gives 12. One coin balances at 12 cm. Measured: 11.5 cm. Off by 0.5 cm.
Left 2 coins at 6 cm gives 12. Same answer, 12 cm. Measured 12.5 cm. Off by 0.5 cm.
Left 3 coins at 5 cm gives 15. One coin would need 15 cm, which is the end of the ruler. Move it to the end and it just balances.

Your balance diary

Look for balances at home: a door, a scale, a seesaw, a wheelbarrow, scissors. Draw one, and say which side is the long distance.

Puzzle 4.6: Two on the left

Left: 3 coins at 10 cm. Right: 2 coins. Where do they go? _____ cm

Puzzle 4.7: Too far

Left: 5 coins at 6 cm. You have one coin for the right. Where would it need to be? _____ cm
Can that work on a 30 cm ruler with the pivot in the middle? _____
What could you do instead? _____

Puzzle 4.8: Equal pairs

Left: 4 coins at 3 cm. Right: 3 coins at 4 cm. Does it balance? _____
Show the numbers: _____

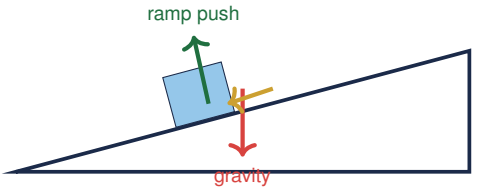
Puzzle 4.9: Mystery bag, again

A bag sits 6 cm left of the middle. Three coins sit 8 cm right of the middle, and the ruler balances. How many coins does the bag weigh? _____

5 Ramps and Effort

A longer, gentler push for a smaller effort

Lifting a heavy box straight up onto a truck is hard. Sliding it up a long plank is easier. The plank is a **ramp**. A ramp does not make the box lighter. It lets you push a little at a time over a longer path.



Try it: How far does the car roll?

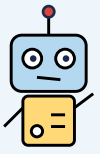
You need: a board, three books, a toy car, a ruler or tape, a pencil.

1. Lean the board on one book. Let the car go from the top mark without pushing. Measure how far from the bottom of the ramp it stops. Do it three times.
2. Repeat with two books, then three books. Same start mark, same car, same floor.
3. Average each set.

Books	Roll 1 (cm)	Roll 2 (cm)	Roll 3 (cm)	Average (cm)
1				
2				
3				

Practice numbers, not from a real test

One book: 62, 60, 64. Total 186, average $186 \div 3 = 62$ cm, spread 4 cm.
Two books: 98, 102, 100. Total 300, average 100 cm, spread 4 cm.
Three books: 130, 132, 134. Total 396, average 132 cm, spread 4 cm.
A steeper ramp sent the car farther each time. The gaps between averages (38 and 32 cm) are much bigger than the spread (4 cm), so the pattern is not just wobble.



The car went farther on the steeper ramp, so a steeper ramp means a bigger force of gravity on the car. Gravity got stronger! Did it?

What makes this correction trustworthy?

What we checked. We changed only the number of books, kept the start mark, the car and the floor the same, and rolled three times for each ramp.

What this check gives us reason to trust. The averages rose with each book and the gaps were much larger than the spread of repeats.

What it does not show. Whether it holds for a heavier car, a rough floor or a much steeper ramp. We tried three heights.

What did we keep?

We drew the car on the ramp as a few arrows. **We kept** _____ **and we set aside** _____.

Pull to lift, or pull up a ramp

Hang a small bag of coins from a rubber band. Lift it straight up and see how far the band stretches. Then pull the same bag up a ramp with the band. The stretch is a stand-in for how hard you pull.

Try it: Straight up or up the ramp?

You need: a rubber band, a bag of coins, a board, books, a ruler.

1. Lift the bag slowly straight up. Measure the stretch while it moves.
2. Lean the board to the same height. Pull the bag slowly up the board. Measure the stretch.
3. Use a longer board to reach the same height. Measure again.

Way up	Stretch (cm)
Straight up	
Short ramp	
Long ramp	

Practice numbers

Straight lift: 6.0 cm stretch. A 40 cm ramp: 3.8 cm. An 80 cm ramp: 2.5 cm.
Longer ramp, smaller pull. But the bag travels 80 cm instead of 40 cm, so the pull lasts twice as far.



kept: the direction of each pull. **lost:** the real shape of the board.

What makes this correction trustworthy?

What we checked. We pulled the same bag the same height three ways and read the band each time.

What this check gives us reason to trust. The stretch got smaller as the ramp got longer: 6.0, 3.8, 2.5.

What it does not show. That the pattern is exact. Friction on the board also changes the pull, and we measured once for each.

What did we keep?

A ramp trades push for distance. **We kept** _____ **and we set aside** _____.

Puzzle 5.1: Average the rolls

A car rolled 55, 57 and 59 cm on a ramp with one book.
Total _____ Average _____ cm

Puzzle 5.2: Predict the roll

On two books the car averaged 100 cm. On three books it averaged 132 cm. Without testing, guess the average for four books.
My guess: _____ cm. Why is it only a guess? _____

Puzzle 5.3: Which ramp?

You must lift a box 1 m onto a truck. One ramp is 2 m long and one is 4 m long. Which needs a smaller push? _____
What do you pay for the smaller push? _____

Puzzle 5.4: Unfair ramp test

Ana rolled a car down a steep ramp on a carpet. Ben rolled the same car down a gentle ramp on a smooth floor. Ben's car went farther. Ana says "a gentle ramp is better". What is wrong with the test?

_____

Puzzle 5.5: Draw the arrows

A sled sits still on a snowy slope. Draw three arrows from the dot: gravity, the slope's push, and friction.



Is anything missing? _____

Puzzle 5.6: Slow down

A toy car rolls down a ramp and then across a rug. On the ramp it speeds up. On the rug it slows. Which arrow gets bigger on the rug? _____

Puzzle 5.7: Gaps

Averages for one, two and three books were 62, 100 and 132 cm. Find the gap from one to two books _____ and from two to three _____.

Puzzle 5.8: How much easier?

Straight lift: 6.0. 40 cm ramp: 3.8. 80 cm ramp: 2.5. By how much smaller is the pull on the 40 cm ramp than the lift? _____
On the 80 cm ramp? _____

Puzzle 5.9: Speeding up

A toy car rolls down a steep ramp, then a gentle ramp. On which does it speed up more? _____
Which arrow along the slope is bigger on the steep one? _____

Puzzle 5.10: Ramp hunt

Find three ramps near you. Write where each is and who it is for.

_____

_____

_____

6 Finding the Missing Arrow

Using leftover arrows to explain what a thing does

A force diagram is a model. It keeps the direction and the relative size of each force, and it drops the exact shape of the thing, its colour and its parts. It is good when the arrows explain what the thing did. If they do not, an arrow is missing.

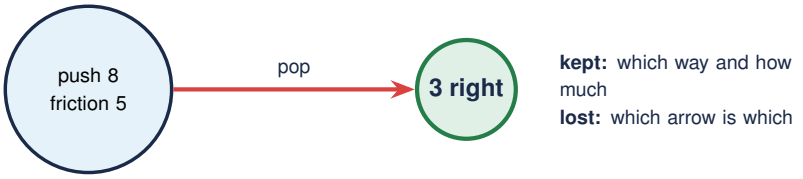
Model	What it keeps	What it drops
An arrow for a force	direction and size	the exact shape of the thing
Popping the arrows	the leftover push	which arrow came from where
Friction arrow	points against sliding	how rough each bump is
Coins times distance	when a seesaw balances	the weight of the ruler
A ramp picture	how steep and how long	the bumps on the board

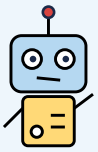
Three rules for reading a diagram

1. Arrows that are equal and opposite pop to nothing. The thing keeps doing what it was doing: staying still, or moving at a steady speed.
2. A leftover arrow means the thing speeds up, slows down or turns, in the direction of the leftover.
3. If the thing does something the arrows cannot explain, look for the missing arrow.

A crate on a floor (practice numbers)

You push a crate right with an arrow of size 8. Friction pulls left with size 5. Pop them: $8 - 5 = 3$ to the right. The crate speeds up, slowly.
Now push with 5. Pop: $5 - 5 = 0$. The crate slides at a steady speed, or stays put.





A sled glides along flat ice at a steady speed, so it must have a leftover push forward to keep it going. Does it?

What makes this correction trustworthy?

What we checked. We checked each diagram by asking whether its arrows explain what the thing did.

What this check gives us reason to trust. When the arrows popped to the same result we saw, the diagram explained it. When they did not, we found the arrow that was missing.

What it does not show. That every force is shown. A diagram can pass on one test and still leave out a small force, such as air.

What did we keep?

We turned a sliding crate into two arrows. **We kept** _____ **and we set aside** _____.

Puzzle 6.1: Spot the missing arrow

A box slides across a floor at a steady speed. Only one arrow is drawn: push right, size 4.



Which arrow is missing, which way does it point, and how big is it? _____

Puzzle 6.2: A ball on a table

A ball sits still on a table. Only gravity is drawn, pointing down, size 3. What is missing?

_____

How big is it? _____

Puzzle 6.3: Pop them

Pop these arrows and write the leftover.

(a) 7 right, 3 left, 2 left: _____

(b) 5 up, 5 down: _____

(c) 4 right, 9 left: _____

Puzzle 6.4: Tug of war

Three children pull a rope left, each with size 4. Two children pull it right, each with size 5. Which way does the rope go, and with what leftover? _____

Puzzle 6.5: The floating boat

A toy boat floats still. Gravity pulls down with size 10. How big is the water's push, and which way? _____

Puzzle 6.6: Choose the surface

From Chapter 2, a block slid on carpet 19 cm, on card 46 cm and on a plastic tray 78 cm. You want a sled to stop soonest. Which surface? _____
Why does it stop soonest there? _____

Puzzle 6.7: A seesaw mystery

A ruler seesaw balances with 4 coins at 5 cm on the left and a bag on the right at 10 cm. How many coins does the bag weigh? _____

Puzzle 6.8: Skater

A skater glides at a steady speed on flat ice. How many forward arrows are needed to explain this? _____
Name the two arrows that do pop to nothing, one up and one down. _____

Puzzle 6.9: Arrows for a leaf

A leaf drifts down at a steady speed. Gravity pulls down with size 2. How big is the air's push, and which way? _____

Puzzle 6.10: Pop them again

9 right, 4 left, 5 left. Leftover: _____

Puzzle 6.11: The speeding car

A toy car rolls down a ramp. Gravity pulls down and the ramp pushes out. A small leftover points down the slope. Does the car speed up, slow down or stay steady? _____

Puzzle 6.12: Another seesaw

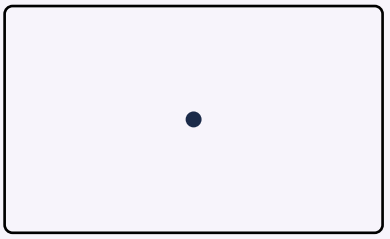
Left: 2 coins at 12 cm. Right: 4 coins at 6 cm. Does it balance? _____ Show the numbers. _____

Puzzle 6.13: A neat diagram

Ben says his force diagram must be right because it looks neat. What should he check instead? _____

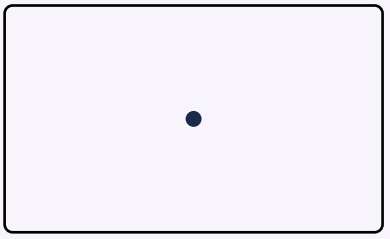
Puzzle 6.14: Draw the arrows: a parachute

A toy parachute drifts down at a steady speed. Draw the arrows from the dot and label each size.



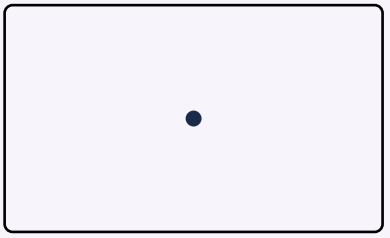
Puzzle 6.15: Draw the arrows: a speeding box

A box is pushed right with size 7 across a floor. Friction is size 3. Draw the horizontal arrows and write the leftover. _____



Puzzle 6.16: Draw the arrows: a floating boat

A boat floats still on a pond. Draw the arrows from the dot and say why they pop to nothing. _____

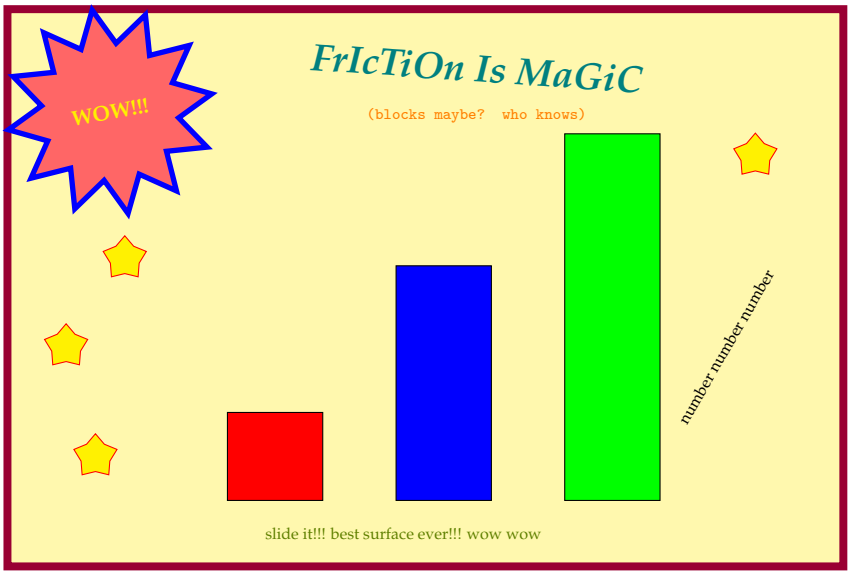


Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about sliding on different surfaces. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so the bars cannot be read.

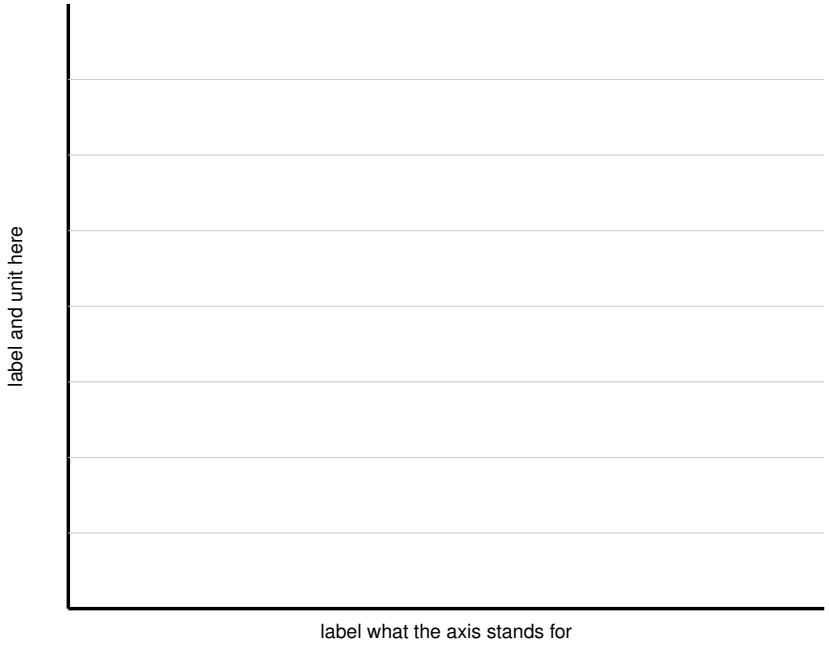
_____

(d) Which part is not relevant to sliding at all?

_____

Now you try

On a separate sheet, redraw a clear bar chart of how far the block slid: carpet 19 cm, card 46 cm and plastic tray 78 cm. Give it a title, label the axes with units, use one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.

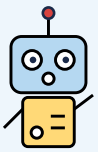


Title:

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Pushes and pulls	a push drawn as an arrow	direction and size	the shape of the thing
2 Sliding			
3 Falling and floating			
4 Balance			
5 Ramps			
6 Missing arrow			



Everything you drew in this book was a simplification of something. A model is allowed to drop things, as long as you say which. Next time something stops, you will know to look for the arrow.

Write a force report

1. **What I looked at.**
2. **What I changed, measured and kept the same.**
3. **My readings.** All of them.
4. **My answer.** With averages and spread.
5. **What I kept and what it hides.**

Pick one test from this book and write a report on it. Use the lines below.

Tests to try

Pick one and design it with the habits from this book: change one thing, keep the rest the same, repeat, and average.

1. Does the weight of a block change how far it slides down a ramp?
2. Does a wider or longer sheet of paper fall more slowly?
3. Does the length of the pull string matter for the rubber-band stretch?
4. What changes how many coins a clay boat can hold?
5. Does a heavier car roll farther down the same ramp?
6. Does the pivot's position change where a ruler balances?

Write the test you pick:

_____

_____

_____

_____

_____

_____

Change: _____ Measure: _____ Keep the same: _____

A week of forces

Each day, notice one push or pull and draw its arrows.

Day	What pushed or pulled	Arrows I would draw
Mon		
Tue		
Wed		
Thu		
Fri		
Sat		
Sun		

Questions I still have

Write what you still wonder about forces. Then pick one and think how to test it.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on stillness. No. A book on a table has two arrows, gravity down and the table's push up, and they pop to nothing. **Keep box.** Kept: the direction and the size of each push. Set aside: the exact shape of the thing.

1.1 Gravity pulls the bag down. The string pulls the bag up.
1.2 (a) $6 - 2 = 4$ right. (b) 0. (c) $5 - 3 = 2$ left.
1.3 Answers depend on the diagram; check that the leftover points the way the thing moves.
1.4 Gravity down, the floor's push up, same size.
1.5 Push or pull, by your own check: a door handle is a pull to open it, a hand on a wall is a push.
1.6 Another arrow of size 2 to the right. **1.7** 1 to the right. **1.8** $7 - 4 = 3$ to the right, so the boat goes right.
1.9 Answers vary.

Chapter 2

Multix on friction. No. The stretch of the rubber band is a stand-in for the pull, and 3 cm is a measure of the pull, not a force in other units. **Keep box.** Kept: friction as one arrow against sliding. Set aside: every bump.

2.1 Rug: $22 + 25 + 23 = 70$, average $70 \div 3 \approx 23$ cm. Floor: $62 + 58 + 60 = 180$, average 60 cm.
2.2 Carpet stops it soonest. Its slide distance is the smallest.
2.3 Look for two changes at once, such as a different push and a different surface.
2.4 Friction helps shoes grip, brakes stop wheels, and knots hold.
2.5 A steady pull means the pull and friction are equal.
2.6 Card has more friction. $3.1 - 1.9 = 1.2$ cm. **2.7** About 12 cm (11 to 13 is fine). It is a guess because we did not test four blocks. **2.8** Left, then right: friction points against the sliding. **2.9** Answers vary.

Chapter 3

Multix on falling. The two do not fall together because the air pushes up more on the flat paper. **Keep box.** Kept: gravity down and an air push up. Set aside: the swirls in the air.

3.1 Crumpled: $0.6 + 0.7 + 0.6 + 0.8 + 0.6 = 3.3$, average $3.3 \div 5 = 0.66 \approx 0.7$ s. Flat: $1.9 + 2.2 + 2.0 + 2.1 + 1.8 = 10.0$, average 2.0 s.
3.2 Gravity down, air push up, nearly equal.
3.3 Gravity down, air push up, same size in the middle of the fall. A heavy flat boat can float because a wide boat gets pushed up by more water.
3.4 Marco changed the paper and the drop height. He should change only one.
3.5 $2.0 - 0.7 = 1.3$ s. About 3 times as long.
3.6 The fall time gets longer, because the air's push up is bigger and the leftover is smaller. **3.7** 2 times. A wider boat pushes aside more water, so the water pushes up more. **3.8** $5 - 1 = 4$ down. Later the air's push is 5.

Chapter 4

Multix on the seesaw. No. $3 \times 4 = 12$ and $1 \times 4 = 4$. They are not equal, so the left side goes down. **Keep box.** Kept: coins times distance. Set aside: the weight of the ruler.

4.1 $2 \times 9 = 18$, and $3 \times 6 = 18$, so 6 cm.
4.2 Left: $3 \times 5 = 15$. Right: $2 \times 8 = 16$. The right side goes down.

4.3 $5 \times n = 2 \times 10 = 20$, so $n = 4$ coins.
4.4 $30 \times 2 = 60$ and $40 \times 1.5 = 60$, so 1.5 m.
4.6 $3 \times 10 = 30$ and $30 \div 2 = 15$ cm. **4.7** $5 \times 6 = 30$, so 30 cm, which is past the end of a 30 cm ruler. Use two coins at 15 cm. **4.8** $4 \times 3 = 12$ and $3 \times 4 = 12$. Yes. **4.9** $3 \times 8 = 24$ and $24 \div 6 = 4$ coins.
4.5 $2 + 6 = 8$ and $1 + 12 = 13$, so the rule "plus" does not work. Coins times distance does.

Chapter 5

Multix on the ramp. No. Gravity is the same. A steeper ramp lets more of the pull act along the slope. **Keep box.** Kept: the directions of the forces. Set aside: the shape of the board.

5.1 $55 + 57 + 59 = 171$, average $171 \div 3 = 57$ cm.
5.2 Any guess above 132 is fine, for example 160. It is only a guess, because we did not test four books.
5.3 The 4 m ramp needs a smaller push, but you push it twice as far.
5.4 Two things changed, the ramp and the floor.
5.5 Gravity down, the slope's push out from the slope, and friction up the slope. Nothing is missing.
5.6 Friction.
5.7 $100 - 62 = 38$ and $132 - 100 = 32$. **5.8** $6.0 - 3.8 = 2.2$ and $6.0 - 2.5 = 3.5$. **5.9** The steep one; the arrow along the slope is bigger. **5.10** Answers vary.

Chapter 6

Multix on the sled. No. The steady sled has no leftover. Friction is small, so the arrows pop to nearly nothing. **Keep box.** Kept: direction and size. Set aside: which arrow is which.

6.1 Friction, pointing left, size 4.
6.2 The table's push, up, size 3.
6.3 (a) $7 - 3 - 2 = 2$ right. (b) 0. (c) $9 - 4 = 5$ left.
6.4 Left: $3 \times 4 = 12$. Right: $2 \times 5 = 10$. Left wins by 2.
6.5 Water pushes up with size 10.
6.6 Carpet. It has the most friction.
6.7 $4 \times 5 = 20$ and $2 \times 10 = 20$, so 2 coins. Check: if the bag is at 10 cm and the coins at 5 cm, the bag is 2 coins.
6.8 None. The ice pushes up and gravity pulls down.
6.9 Air pushes up with size 2.
6.10 0. **6.11** It speeds up. **6.12** $2 \times 12 = 24$ and $4 \times 6 = 24$. Yes. **6.13** Check that the arrows explain what the thing did. **6.14** Gravity down 2, air up 2. **6.15** $7 - 3 = 4$ right. **6.16** Gravity down and water up, equal.
Slop check. (a) You cannot tell: no title, no labels. (b) Many colours, stars, joke text, tilted filler. (c) A title, the surfaces and the distances with units. (d) The stars and jokes.

Words

balance when the turning effects on both sides are equal.

force a push or a pull.

friction a force that points against sliding.

gravity the pull that draws things toward the Earth.

leftover what is left when the arrows are popped.

model a simple stand-in that keeps some things and drops others.

pivot the point a seesaw balances on.

ramp a slope used to lift things gently.

spread how far apart repeated readings are.

Notes for grown-ups

This book teaches that a force diagram is a model: it reduces a push or pull to an arrow with a direction and a size. Each chapter has one activity, one worked example with made-up practice numbers, one trust box, one "what did we keep?" prompt and a page of puzzles.

Materials and safety. A ruler, a pencil, ten coins, rubber bands, a board, books, a toy car and string. Keep ramps low and away from stairs. Do not drop things from windows or stairs. Rubber bands can snap: do not stretch them toward faces.

The practice numbers were made up so the arithmetic is easy to check. Real results depend on the surfaces and the objects, so real results will differ.

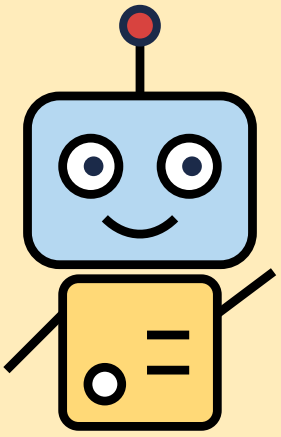
Sources

- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For repeats and spread.
- D. Halliday, R. Resnick and J. Walker. *Fundamentals of Physics*. Wiley. For forces, friction and balance.

Next in Science

Where the Water Goes

A cup of water sits on a shelf, and a day later there is less. Book 5 follows water as it changes and moves, and keeps track of how much there was.



Ages 9 to 11 ⋆ **Science Book 4**